

Forecasting Next-Quarter Revenue for U.S. Public Companies Using SEC Filings and Random Forest Models

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Abstract

This project develops and evaluates a firm-level system for forecasting next-quarter revenue for U.S. public companies. The primary data source is the U.S. Securities and Exchange Commission Financial Statement Data Sets, supplemented with Consumer Price Index, federal funds rate, and unemployment data from Federal Reserve Economic Data. The workflow includes revenue extraction from heterogeneous SEC filings, fiscal-quarter normalization, derived fourth-quarter revenue, filing-timing validation, chronological model evaluation, benchmark comparison, and sensitivity analysis. The initial directly reported SEC revenue panel contained 67,094 observations. Adding derived fiscal-Q4 observations produced an enhanced panel of 84,108 observations. After next-quarter target and feature construction, the candidate modeling dataset contained 51,168 firm-quarter observations. A filing-timing audit excluded 2,366 observations (4.62%) because the selected filing date was not earlier than the target-period date. The final timing-valid dataset contained 48,802 observations from 4,528 firms. The final model was a Random Forest trained to predict next-quarter revenue growth in log space. After lag construction, model development used target quarters from 2022Q2 through 2023Q4 for training and 2024 for validation. Final performance was measured on a held-out 2025 test set containing 12,924 firm-quarter observations from 3,586 firms. The final model achieved a mean absolute error of \$153.36 million compared with \$191.94 million for the seasonal baseline, a 20.10% reduction. Median absolute percentage error was 7.31%, and sMAPE was 0.18 among observations with positive actual revenue. Of those observations, 58.8% were within $\pm 10\%$ of actual revenue and 76.7% were within $\pm 20\%$. The final model produced no negative forecasts. Sensitivity analysis identified a specific weakness: the model reduced MAE by 27.38% relative to the seasonal baseline on directly reported quarters but had 8.83% higher MAE on derived fiscal-Q4 observations. A controlled macro-feature ablation showed that the six macroeconomic features improved MAE by only 0.98%, indicating a small observed incremental contribution on this test set.

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1. Introduction

1.1 Problem and Research Question

Revenue forecasting is a basic planning problem in corporate finance. A forecast of next-quarter revenue can support budgeting, resource allocation, management review, and variance analysis. The forecasting problem is also suitable for data science because public-company revenue is reported repeatedly over time and can be combined with firm characteristics and broader economic indicators.

The task is not simply to fit historical revenue as closely as possible. A useful forecasting model must operate on information that would have been available before the outcome being predicted. It also needs to be compared with simple forecasting rules that are already difficult to beat. Quarterly revenue often has persistence and seasonality, so a model that appears accurate in isolation may provide little benefit over using the previous quarter or the same quarter of the previous year.

The research question is: Can historical SEC revenue data, together with firm-level and macroeconomic features, improve next-quarter revenue forecasts for U.S. public companies relative to simple quarterly forecasting baselines?

The project therefore evaluates more than model accuracy alone. Data timing, fiscal-Q4 construction, benchmark performance, forecast validity, and subgroup sensitivity are treated as part of the forecasting problem.

1.2 Scope and Objectives

The unit of observation is a firm-quarter. Each modeling row contains revenue and other information for a current observation period and a target containing revenue for the following quarter. The project forecasts quarterly revenue only. It does not forecast expenses, cash flow, earnings, or complete financial statements, and it does not estimate causal effects of inflation, interest rates, or unemployment on company revenue.

The project has four practical objectives: construct a consistent revenue panel from heterogeneous SEC filings; forecast next-quarter revenue with chronological out-of-sample evaluation; prevent information leakage and economically invalid predictions; and produce outputs that can support forecast-versus-actual review.

2. Data Description

2.1 Public Data Sources

The primary financial data source was the SEC Financial Statement Data Sets (U.S. Securities and Exchange Commission, n.d.). These datasets contain structured numeric information from public-company filings and associated submission, tag, and presentation metadata. The project used the SEC data to obtain company identifiers, filing dates, forms, fiscal periods, SIC codes, and reported revenue values.

Three macroeconomic series were obtained through Federal Reserve Economic Data (FRED): the Consumer Price Index for All Urban Consumers (CPIAUCSL), the Federal Funds Effective Rate (FEDFUNDS), and the U.S. unemployment rate (UNRATE) (Board of Governors of the Federal Reserve System (US), n.d.; U.S. Bureau of Labor Statistics, n.d.-a, n.d.-b). The series were aligned to quarters and used as economy-wide context rather than as causal variables.

2.2 SEC Revenue Extraction

The SEC data do not provide one universal field that can be selected as revenue for every company. Firms use different XBRL tags, and the same filing may contain multiple candidate revenue values. The project therefore applied a priority rule to select one revenue concept:

RevenueFromContractWithCustomerExcludingAssessedTax, Revenues,
RevenueFromContractWithCustomerIncludingAssessedTax, and Revenue, in that order.

Candidate observations were filtered for income-statement revenue, quarterly reporting, USD denomination, consolidated company values, valid forms, and the current reporting period. When multiple candidate rows remained within a filing, the higher-priority revenue tag was retained. When multiple filings existed for a company-period, a deterministic filing-selection rule was applied. The resulting directly reported SEC revenue panel contained 67,094 observations.

2.3 Fiscal-Q4 Derivation

Many annual 10-K filings report full-year revenue without a separate discrete fourth-quarter value. Removing these cases would reduce coverage and make Q4 underrepresented. When annual revenue and Q1-Q3 values were available, fiscal-Q4 revenue was calculated as annual revenue minus Q1, Q2, and Q3. Each derived observation was flagged with `is_derived_q4 = 1` and its derivation source was retained. Adding derived fiscal-Q4 observations expanded the revenue panel from 67,094 to 84,108 observations.

Derived Q4 improves panel completeness but can propagate errors from the annual value or any of the first three quarters. The final evaluation therefore treats derived-Q4 observations as a separate sensitivity group rather than assuming they are equivalent to directly reported quarters.

2.4 Modeling Dataset and Chronological Splits

The enhanced revenue panel was transformed into a next-quarter forecasting dataset. Each row represents a current firm-quarter and its next-quarter revenue target. Target construction and feature requirements reduced the enhanced panel to 51,168 candidate modeling observations before the timing audit.

The modeling split is keyed to the target quarter. After lag construction, the training targets span 2022Q2 through 2023Q4, validation uses 2024 targets, and final evaluation uses 2025 targets. This prevents later target periods from being mixed into model development through a random split.

Table 1. Final modeling sample after timing validation

Split	Final observations	Unique firms	Derived-Q4 rows	Direct rows
Training, targets 2022Q2-2023Q4	22,559	4,068	2,546	20,013
Validation, 2024 targets	13,319	3,714	2,726	10,593
Test, 2025 targets	12,924	3,586	2,709	10,215
Total	48,802	4,528	7,981	40,821

2.5 Filing-Timing Audit

A final audit identified a point-in-time issue. A historical filing should not be used as a predictor if the selected filing had not been filed before the target-period date. The project therefore defined `forecast_timing_valid` as `filed_date < target_period_date`. This rule represents forecasting the next reported quarter using filing information already available before the target-period date; it should not be interpreted as a strict forecast made on the first day of the target quarter.

Of 51,168 candidate modeling observations, 2,366 (4.62%) failed this timing rule and were excluded, leaving 48,802 observations. The exclusions were not evenly distributed across the model splits or reporting categories.

Table 2. Timing-validity exclusions by model split

Split	Candidate rows	Excluded	Exclusion rate	Final rows
Training, targets 2022Q2-2023Q4	23,547	988	4.20%	22,559
Validation 2024	14,125	806	5.71%	13,319
Test 2025	13,496	572	4.24%	12,924
Total	51,168	2,366	4.62%	48,802

Exclusion rates were calculated by dividing the number of excluded observations by the number of candidate observations within the same group. When reporting a group's share of all timing exclusions, the denominator was the 2,366 excluding observations.

Q1 targets were disproportionately affected. Of 10,188 candidate Q1 observations, 1,336 were excluded, producing an exclusion rate of 13.11%. These 1,336 observations accounted for 56.47% of all 2,366 timing exclusions.

Derived fiscal-Q4 observations also failed the timing rule more often. Of 9,409 candidate derived-Q4 observations, 1,428 were excluded, producing an exclusion rate of 15.18%. In comparison, 938 of 41,759 directly reported observations were excluded, producing an exclusion rate of 2.25%. These groups overlap with the quarterly results: 1,231 of the 1,336 timing-invalid Q1 target observations used a derived-Q4 current-quarter value.

Amended filings were another concentrated source of timing-invalid observations. The audit excluded 347 of 454 10-Q/A observations, an exclusion rate of 76.43%, and 223 of 254 10-K/A observations, an exclusion rate of 87.80%. Combined, 570 of 708 amended-filing observations were excluded, producing an exclusion rate of 80.51%. These 570 observations represented 24.09% of all 2,366 timing-invalid observations.

Because the Q1, derived-Q4, and amended-filing categories overlap, their counts should not be added together.

3. Methods and Algorithms

3.1 Forecasting Design

The project treats next-quarter revenue forecasting as a supervised regression problem. Model fitting uses earlier target periods and final accuracy is measured on a later held-out year. After lag construction, training targets span 2022Q2 through 2023Q4, validation uses 2024, and the 2025 test period is reserved for final evaluation. This follows the general forecasting principle that accuracy should be measured on observations not used to fit or select the model (Hyndman & Athanasopoulos, 2021).

3.2 Feature Engineering

The saved final model uses 22 features. Seven variables describe firm-level revenue level and growth: signed log current revenue, signed-log revenue lags at one and four quarters, signed-log revenue changes at one and four quarters, and revenue-growth lags at one and four quarters. Six macroeconomic variables provide economy-wide context: CPI, the federal funds rate, unemployment, and quarter-over-quarter changes in each. Nine one-hot sector indicators are derived from SIC classifications.

The model therefore combines firm-specific temporal information, macroeconomic context, and broad sector membership. Feature importance is not used as a primary result because the project question concerns out-of-sample forecasting performance rather than causal interpretation of individual variables.

3.3 Forecasting Baselines

Two simple benchmarks were retained in the final evaluation. The seasonal naive forecast for a target quarter equals revenue from the same fiscal quarter one year earlier, providing a strong reference when annual seasonality is present. The last-quarter naive forecast equals current-quarter revenue and tests whether a more complex model improves on simple persistence.

3.4 Development-Stage Residual-Correction Models

An earlier project stage attempted to improve the year-over-year baseline by predicting its raw-dollar residual and adding the predicted residual back to the baseline. That approach demonstrated why model selection could not rely on dollar MAE alone. A development-stage Random Forest residual model produced 1,836 negative forecasts. Those earlier residual-model results were measured on a 13,496-row pre-audit test panel and are not included in the final benchmark table because the final timing-valid test sample contains 12,924 rows.

3.5 Final Log-Growth Random Forest

The final approach predicts next-quarter proportional revenue change rather than raw dollar revenue. For positive current and next-quarter revenue in the historical training sample, the target is $\log(\text{next-quarter revenue} / \text{current-quarter revenue})$. The forecast is reconstructed as current revenue multiplied by $\exp(\text{predicted log growth})$. Modeling growth reduces the tendency of a single model to focus mainly on differences in firm scale.

The fitted estimator is a RandomForestRegressor with 500 trees, minimum leaf size 20, random seed 0, and model version final_v2_timing_valid. The serialized final model confirms these settings. Random Forests aggregate predictions across randomized decision trees and can model nonlinear relationships and interactions without requiring a single linear functional form (Breiman, 2001). The implementation used scikit-learn (Pedregosa et al., 2011).

Hyperparameters were selected using only the 2024 validation split, with validation sMAPE as the selection criterion. The 500-tree, minimum-leaf-size-20 specification was locked before the held-out 2025 test set was evaluated; the test set was not used for model or hyperparameter selection.

The timing-valid training split contains 22,559 rows, of which 21,781 were usable as positive current-and-target revenue pairs for the log-growth fit. At prediction time, routing depends only on current revenue: rows with positive current revenue use the Random Forest prediction, while rows with non-positive current revenue use the seasonal-naive fallback. Final forecasts are clipped at zero. This rule does not inspect the future target and therefore does not create target leakage. The 2025 test set contains 332 fallback observations.

3.6 Evaluation Metrics

No single error measure is sufficient because firms differ greatly in revenue scale. MAE reports average absolute dollar error. RMSE gives greater weight to large misses. Median absolute percentage error describes a typical proportional miss, while sMAPE provides a symmetric percentage-based error measure and is calculated as the mean of $2|\text{forecast} - \text{actual}| / (|\text{actual}| + |\text{forecast}|)$. It is reported on a 0-2 decimal scale, so 0.182 corresponds to 18.2%. The report also uses hit rates within $\pm 5\%$, $\pm 10\%$, and $\pm 20\%$ of actual revenue. Forecasting research has documented limitations of percentage error measures when actual values are zero or otherwise problematic (Hyndman & Koehler, 2006).

MAE and RMSE use all 12,924 final test observations. Median APE, sMAPE, and hit-rate measures are calculated only on the 12,577 observations with positive actual revenue. The remaining 347 observations have non-positive actual revenue and are excluded only from percentage-based metrics.

3.7 Derived-Q4 Sensitivity Test

A sensitivity analysis evaluates the same final forecasts separately according to whether the current-quarter revenue feature was directly reported or derived fiscal Q4. The 2025 test set contains 10,215 directly reported observations (79.04%) and 2,709 derived-Q4 observations (20.96%). No separate model is fitted for this analysis.

3.8 Macroeconomic Feature Ablation

A controlled ablation test measures whether the six macroeconomic variables improve the final forecasting model. The saved 22-feature model is compared with an otherwise identical 16-feature Random Forest that removes CPI, the federal funds rate, unemployment, and their quarter-over-quarter changes. Both models use the same 21,781 training rows, the same 12,924-row 2025 test set, 500 trees, minimum leaf size 20, random seed 0, fallback rule, and metric definitions. The only experimental difference is the presence or absence of the six macro features.

4. Results

4.1 Final Test Sample

The locked final evaluation contains 12,924 2025 firm-quarter observations from 3,586 firms. Of these, 12,577 observations have positive actual revenue and are included in percentage-based metrics. All three final benchmark methods are evaluated on the same 12,924 observations.

4.2 Baseline and Final Model Performance

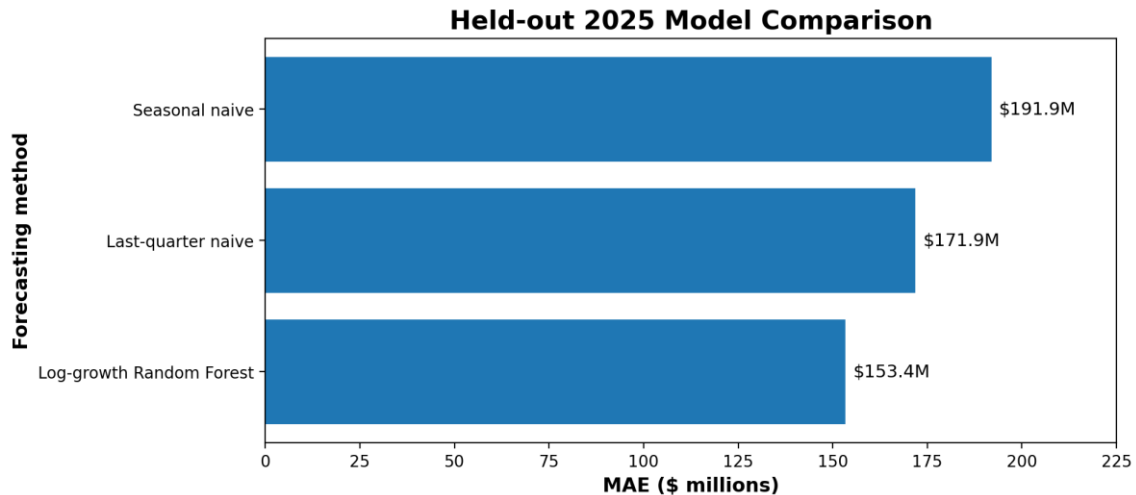
Table 3. Held-out 2025 forecast performance

Metric	Seasonal naive	Last-quarter naive	Log-growth RF
Test observations	12,924	12,924	12,924
Test firms	3,586	3,586	3,586
MAE (\$M)	191.94	171.85	153.36
RMSE (\$M)	1,182.73	1,258.07	1,018.51
Median APE	10.63%	7.34%	7.31%
sMAPE	0.251	0.189	0.182
Within $\pm 5\%$	26.78%	38.23%	38.40%
Within $\pm 10\%$	47.80%	59.25%	58.78%
Within $\pm 20\%$	70.62%	76.91%	76.70%
Negative forecasts	64	54	0
Zero forecasts	244	278	203

Note. MAE and RMSE use all 12,924 test observations. Median APE, sMAPE, and hit rates use the 12,577 observations with positive actual revenue. sMAPE is reported on a 0-2 decimal scale ($0.182 = 18.2\%$).

The final Random Forest has the lowest MAE, RMSE, median absolute percentage error, and sMAPE. MAE decreases from \$191.94 million for the seasonal baseline to \$153.36 million, a 20.10% reduction. The final model also produces zero negative forecasts.

The last-quarter baseline remains competitive on threshold-based percentage accuracy. Its $\pm 10\%$ hit rate is 59.25%, slightly above 58.78% for the final model, and its $\pm 20\%$ hit rate is 76.91%, slightly above 76.70%. The final model therefore does not dominate every metric. Its main advantages are lower dollar error, lower sMAPE, slightly lower median percentage error, and elimination of negative forecasts.



Note: Log-growth Random Forest MAE is 20.10% lower than the seasonal naive baseline.

Figure 1. Held-out 2025 MAE by forecasting method. All methods use the same 12,924-observation test set.

4.3 Actual Versus Forecast Revenue

Aggregate error metrics do not show how predictions behave across the wide range of firm sizes. Figure 2 therefore plots all 12,924 test observations as a diagnostic. Because actual revenue includes 347 non-positive values, actual and predicted revenue were floored at zero for this visualization before applying $\log_{10}(1 + \text{revenue})$. This visualization-only transformation does not change any evaluation metric. A 45-degree reference line represents exact predictions.

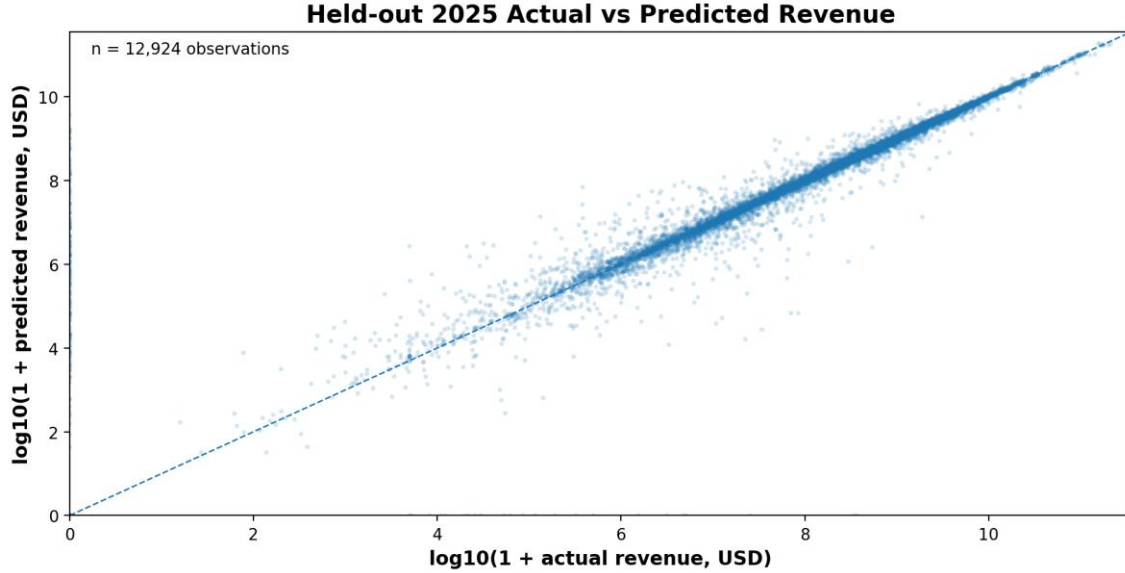


Figure 2. Actual versus predicted 2025 revenue for all 12,924 test observations. Non-positive values are displayed at zero before the $\log_{10}(1 + \text{revenue})$ transformation.

4.4 Distribution of Percentage Accuracy

The final model has a median absolute percentage error of 7.31%. This does not mean that every forecast is close to 7.31%. Among the 12,577 observations with positive actual revenue, 38.40% are within $\pm 5\%$ of actual revenue, 58.78% are within $\pm 10\%$, and 76.70% are within $\pm 20\%$. An empirical cumulative distribution function (ECDF) shows these thresholds while retaining the full distribution of absolute percentage errors.

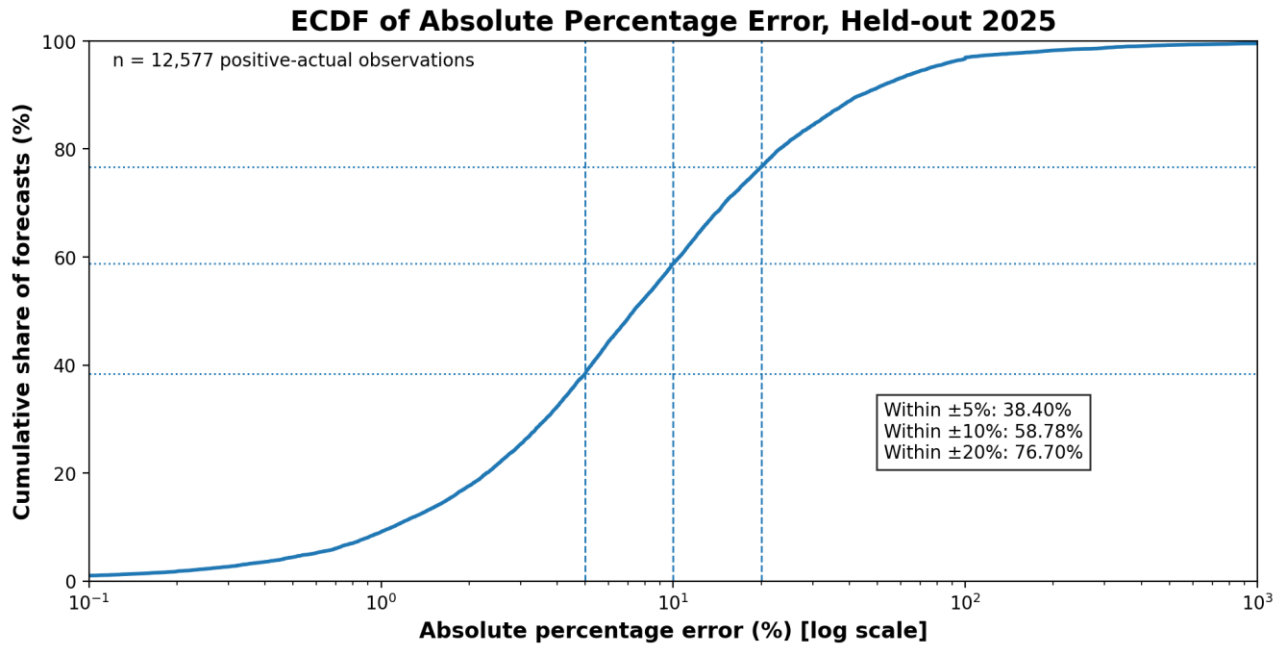


Figure 3. Cumulative distribution of absolute percentage error

The final RMSE is \$1,018.51 million, substantially larger than the MAE of \$153.36 million. This difference indicates a long upper tail in dollar errors: a limited number of large misses have a strong effect on squared error even though the median proportional error is much smaller.

4.5 Derived Fiscal-Q4 Sensitivity

Table 4. Sensitivity to current-quarter revenue source

Metric	Directly reported	Derived fiscal Q4
Test observations	10,215	2,709
Share of test set	79.04%	20.96%
Positive actual observations	9,903	2,674
Final-model MAE (\$M)	140.90	200.36
Final-model RMSE (\$M)	947.82	1,249.59
Median APE	7.05%	8.63%
sMAPE	0.181	0.187
Within $\pm 10\%$	59.89%	54.68%
Within $\pm 20\%$	77.63%	73.26%
Seasonal baseline MAE (\$M)	194.02	184.10
MAE reduction vs seasonal	27.38%	-8.83%
Fallback observations	257	75

The subgroup result is materially different from the overall test result. For directly reported quarters, the final model reduces MAE from \$194.02 million to \$140.90 million, a 27.38% improvement over the seasonal baseline. For derived fiscal Q4, final-model MAE is \$200.36 million compared with \$184.10 million for the seasonal baseline, meaning the final model is 8.83% worse on this subgroup.

Percentage metrics show the same direction. Median APE increases from 7.05% on directly reported observations to 8.63% on derived-Q4 observations, while the $\pm 10\%$ hit rate falls from 59.89% to 54.68%.

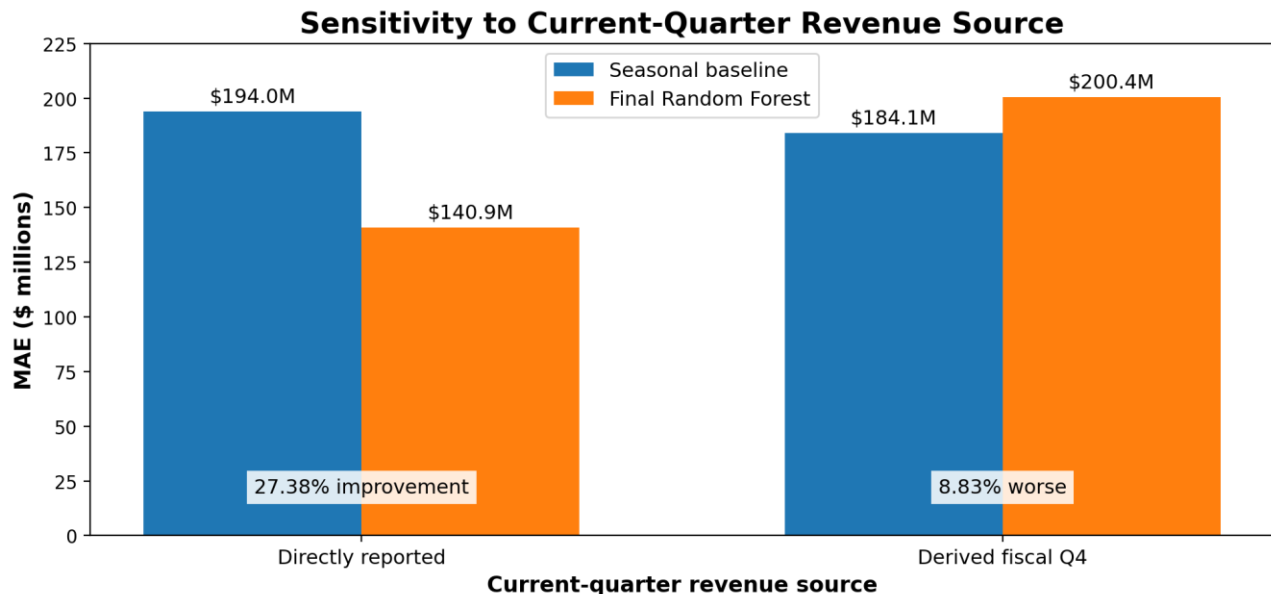


Figure 4. Final-model and seasonal-baseline MAE by current-quarter revenue source.

4.6 Macroeconomic Feature Ablation

Table 5. Controlled macro-feature ablation on the final 2025 test set

Metric	With macro	Without macro
MAE (\$M)	153.36	154.88
RMSE (\$M)	1,018.51	1,007.50
Median APE	7.31%	7.56%
sMAPE	0.182	0.184
Within $\pm 5\%$	38.40%	37.88%
Within $\pm 10\%$	58.78%	58.12%
Within $\pm 20\%$	76.70%	76.66%
Negative forecasts	0	0
Fallback observations	332	332

Removing the six macroeconomic variables increases MAE from \$153.36 million to \$154.88 million. The with-macro model therefore has 0.98% lower MAE. Median APE, sMAPE, and the $\pm 5\%$ and $\pm 10\%$ hit rates also improve modestly when the macro variables are included.

The result is not uniform across every metric. RMSE is lower without macro variables: \$1,007.50 million versus \$1,018.51 million with macro variables. The macro features therefore improve typical and average absolute accuracy slightly but do not reduce the largest dollar errors.

5. Discussion

5.1 Interpretation of the Final Model

The final results support the log-growth Random Forest as the primary model among the methods evaluated on the timing-valid test sample. Its strongest advantage is dollar accuracy: MAE is 20.10% lower than the seasonal baseline, and RMSE is also lower. The model also improves median percentage error and sMAPE while producing no negative forecasts.

The comparison with the last-quarter baseline is more nuanced. The last-quarter rule already performs strongly because quarterly company revenue is persistent. Its $\pm 10\%$ and $\pm 20\%$ hit rates are marginally higher than those of the Random Forest. This means the machine-learning model should not be described as uniformly superior. A simple persistence forecast remains a useful benchmark and could remain visible beside the model in a financial review process.

Percentage-based accuracy measures describe the 12,577 test observations with positive actual revenue, while MAE and RMSE include the full 12,924-observation test set, including fallback cases.

5.2 Why the Log-Growth Formulation Helped

The dataset contains firms with very different revenue scales. A model trained directly on dollar revenue or raw-dollar residuals can devote substantial capacity to learning firm size. The log-growth target instead asks the model to estimate proportional movement relative to the company's current revenue.

This formulation does not eliminate all scale effects, because the final dollar metrics are still larger for companies with larger revenue. It does, however, produce a target that is more comparable across firms. The earlier residual-correction experiment reinforces this point: a lower development-stage dollar MAE did not compensate for 1,836 negative forecasts and poor proportional behavior.

5.3 Contribution of Macroeconomic Features

The controlled ablation provides a direct test of the macro features rather than relying on an exploratory regression. The with-macro and without-macro models use the same training and test observations, hyperparameters, fallback logic, and evaluation metrics. Only the six macro variables differ.

The macroeconomic features show a small observed incremental benefit on this test set. MAE improves by 0.98%, median APE decreases from 7.56% to 7.31%, sMAPE decreases from 0.184 to 0.182, and the $\pm 10\%$ hit rate increases from 58.12% to 58.78%. RMSE, however, is slightly worse with macro variables. The

evidence therefore supports a limited conclusion: CPI, interest rates, and unemployment add some context on this test sample, but firm-specific revenue history remains the main forecasting signal. Robustness across repeated random seeds or alternative test windows was not established.

The project does not use the optional ordinary-least-squares macro regression to claim causal effects or statistical significance. Macro values are repeated across many firms within the same quarter, so a simple firm-row regression is not an appropriate basis for causal interpretation. The final report relies on the controlled forecasting ablation instead.

5.4 Timing Validity and Sample Selection

The timing audit improves the credibility of the evaluation because it removes rows whose selected filing did not satisfy the project's information-availability rule. Without this correction, 2,366 rows would remain in the candidate sample despite a timing inconsistency.

The exclusions are not random. Q1 targets account for 56.47% of all exclusions and have a 13.11% exclusion rate. Derived-Q4 observations have a 15.18% exclusion rate compared with 2.25% for directly reported observations. Amended filings are especially affected: 570 of 708 10-Q/A and 10-K/A observations fail the rule. These patterns overlap, particularly Q1 targets and derived Q4.

The result is a cleaner but more selective dataset. Final metrics describe the timing-valid sample and may overrepresent cleaner filing situations relative to Q1, derived-Q4, and amended-filing cases. Because the timing rule uses the target quarter-end date rather than a fixed forecast-origin date, forecast lead times vary across observations. The macroeconomic series also do not preserve point-in-time vintages or publication lags. The results should therefore be interpreted as a timing-screened historical evaluation rather than a fully real-time forecasting backtest. A stricter future design could preserve full filing histories and reconstruct the exact latest filing and macro values available at a fixed forecast-origin timestamp.

5.5 Derived-Q4 Weakness

Derived fiscal Q4 is the clearest observed subgroup weakness. These observations represent 20.96% of the 2025 test set, and their final-model MAE is \$200.36 million compared with \$140.90 million for directly reported observations. More importantly, the final model beats the seasonal baseline by 27.38% on directly reported quarters but loses to it by 8.83% on derived Q4.

The sensitivity result means that the overall 20.10% MAE reduction should not be generalized uniformly across quarter types. Q4 derivation may add noise because the value is computed from the annual total and Q1-Q3, so measurement or period-alignment problems in any component can affect the derived quarter. The analysis does not isolate one causal explanation, but it demonstrates that derived-Q4 cases require separate monitoring.

5.6 Error Scale and Remaining Scope Limits

Dollar and percentage metrics answer different questions. The final MAE is \$153.36 million while median APE is 7.31%. RMSE is more than six times the MAE, indicating a substantial tail of large dollar errors. The locked final outputs do not contain an error-concentration table by firm-size decile, so the report does not claim a specific percentage of total error attributable to the largest firms.

Two additional scope limits remain. First, one general model is applied across firms from many industries, so industry-specific revenue processes may not be fully captured. Second, the project forecasts quarterly revenue only; it does not forecast expenses, earnings, working capital, or cash flow.

Final forecasts and locked evaluation metrics were also exported to an Excel variance-analysis workbook and a Power BI-ready dataset for review.

6. Future Work

The highest-priority extension is separate handling of derived fiscal Q4. Because the seasonal baseline outperforms the final model on this subgroup, future work should test a Q4-specific model, a seasonal override, or an ensemble that chooses between the two approaches using only information available at forecast time.

A second extension is a stricter point-in-time data architecture. The pipeline could preserve complete filing histories and reconstruct exactly which filing and macro values were available at a fixed forecast origin instead of selecting a final company-period filing and then auditing its date.

Other useful extensions include industry-specific models, prediction intervals, deeper analysis of extreme dollar errors, and richer firm-level variables such as additional financial statement measures or company guidance. Any new data source should follow the same point-in-time controls used for SEC filings.

Finally, the reporting workflow could be automated so new filings update the panel, generate forecasts, calculate forecast-versus-actual variance when the next quarter becomes available, and refresh the reporting layer.

7. Conclusion

This project developed a time-aware system for forecasting next-quarter revenue for U.S. public companies using SEC financial statements, firm revenue history, macroeconomic context, and Random Forest modeling. The final dataset contains 48,802 timing-valid firm-quarter observations after 2,366 candidate rows were excluded because the selected filing did not satisfy the information-timing rule.

On the held-out 2025 test set of 12,924 observations from 3,586 firms, the final Log-Growth Random Forest reduced MAE by 20.10% relative to the seasonal baseline, from \$191.94 million to \$153.36 million, and produced no negative forecasts. The last-quarter baseline remained competitive on percentage hit rates, showing that simple revenue persistence is a strong benchmark and that the machine-learning model does not dominate every measure.

The most important limitation is that performance is not uniform across reporting situations. The final model improved MAE by 27.38% over the seasonal baseline for directly reported quarters but performed 8.83% worse for derived fiscal Q4. Timing exclusions were also concentrated in Q1, derived-Q4, and amended-filing observations. Macro variables added only a modest 0.98% improvement in MAE.

The project therefore supports a specific conclusion: a log-growth Random Forest can improve overall held-out revenue forecast error on the timing-valid sample, but its value depends on careful data construction, realistic information timing, strong benchmark comparisons, and separate evaluation of cases where the underlying SEC revenue measure is less reliable.

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Appendix A. Reproducibility and Final Model Audit

This appendix records the main sample transitions and final saved-model metadata used to verify the report. It is supplementary to the methods section and does not introduce additional results.

Table A1. Main dataset stages

Stage	Observations	Purpose / change
Direct SEC revenue panel	67,094	Directly reported quarterly revenue observations after extraction and cleaning
Enhanced revenue panel	84,108	Adds separately flagged derived fiscal-Q4 observations
Candidate forecasting dataset	51,168	Rows with next-quarter targets and forecasting features before timing audit
Final timing-valid modeling dataset	48,802	Removes 2,366 rows that fail <code>filed_date < target_period_date</code>

The serialized final model file was inspected directly. It stores a scikit-learn RandomForestRegressor and the ordered feature list used for prediction. The saved metadata matches the final report results.

Table A2. Final saved-model metadata

Saved model field	Verified value
Model name	log_growth_random_forest
Model version	final_v2_timing_valid
Estimator	RandomForestRegressor
Number of features	22
n_estimators	500
min_samples_leaf	20
random_state	0
Timing-valid training rows	22,559
Positive-pair training rows used	21,781
Validation rows	13,319
Test rows	12,924
Test firms	3,586
Forecast timing rule	<code>filed_date < target_period_date</code>
Prediction routing	Random Forest when current revenue > 0; seasonal fallback otherwise
Percentage-metric denominator	actual revenue > 0 only

The final test pipeline contains 332 fallback observations. The six macro features in the saved model are CPI, federal funds rate, unemployment rate, and the quarter-over-quarter change in each series. The controlled ablation removed only those six variables and retained the same estimator settings, training rows, test rows, and routing rules.

Appendix B. Selected Modeling Fields and Forecast Rules

The following fields are the main variables needed to interpret the modeling dataset and final forecast outputs. The complete data dictionary and code remain in the project repository.

Table B1. Selected modeling fields

Field	Description
cik	SEC Central Index Key identifying the firm
sic	Standard Industrial Classification code used for broad sector encoding
period_date	Current financial reporting period date
filed_date	Date the SEC filing was submitted
revenue_usd	Current-quarter revenue in U.S. dollars
target_period_date	Period date of the next-quarter target
target_revenue_next_qtr	Observed next-quarter revenue used as the forecasting target
seasonal_naive_forecast_next_qtr	Seasonal baseline forecast used for comparison and fallback
is_derived_q4	Flag indicating whether current-quarter revenue was derived fiscal Q4
model_split	Pipeline labels train_2021_2023, validation_2024, or test_2025; after lag construction, training target quarters span 2022Q2-2023Q4
forecast_timing_valid	True when filed_date < target_period_date
pred_improved	Final routed forecast after Random Forest / fallback selection and non-negative clipping

For positive historical current and target revenue, the training label is $\log(\text{target revenue} / \text{current revenue})$. At prediction time, the model forecast is current revenue multiplied by $\exp(\text{predicted log growth})$. If current revenue is non-positive, the seasonal-naive forecast is used instead. The routed forecast is then clipped at zero. Percentage metrics are calculated only when actual target revenue is positive.

The project code, processing scripts, and supporting files are available in the [GitHub repository](#).